Let the 2D plane be a matrix

$$A \in \mathbb{R}^{M \times N}, \qquad A_{i,j}, \quad i = 1, \ldots, M, \quad j = 1, \ldots, N,$$

where i indexes rows and j indexes columns.

Partition the rows into K consecutive groups

$$\{1, \ldots, M\} = \bigcup_{k=1}^{K} G_k, \qquad G_k = \{r_{k-1} + 1, \ldots, r_k\},$$

with

$$0 = r_0 < r_1 < \cdots < r_K = M.$$

The scan proceeds by groups. For each group G_k, scan its rows in increasing order,

$$i \in G_k,$$

and within each row scan the columns in increasing order,

$$j = 1, \ldots, N.$$

For a single row i, the column average is

$$\mu_i = \frac{1}{N} \sum_{j=1}^{N} A_{i,j}.$$

For a group G_k, the corresponding column-wise average vector is

$$\mu_{k,j} = \frac{1}{|G_k|} \sum_{i \in G_k} A_{i,j}, \qquad j = 1, \ldots, N,$$

where $|G_k| = r_k - r_{k-1}$.

The overall average over all columns in the rows of group G_k is

$$\overline{A}_k = \frac{1}{N\,|G_k|} \sum_{i \in G_k} \sum_{j=1}^{N} A_{i,j} = \frac{1}{|G_k|} \sum_{i \in G_k} \mu_i.$$

Equivalently, as a sequential scan,

$$\overline{A}_k = \frac{1}{|G_k|} \sum_{i=r_{k-1}+1}^{r_k} \left(\frac{1}{N} \sum_{j=1}^{N} A_{i,j} \right), \qquad k = 1, \ldots, K.$$